

Supplementary Material for “Learning-Based Radio Map Reconstruction: DiReM Dataset and RadioBridge Framework”

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S-I. DATA AND SUPERVISION

Colormap inversion. Each configuration folder contains a rendered coverage map and its corresponding CSV export (longitude, latitude, value, dBm, and linear quantities). Because the simulator colormap is auto-scaled independently for each configuration, a global color-to-value mapping is invalid: a lookup table fitted on one configuration and applied to another yields a mean absolute decoding error of 48.66 value units, compared with 2.76 units for configuration-specific calibration. We therefore construct a separate color-to-value lookup table for each configuration by sampling the rendered coverage map at the CSV coordinates and averaging the numerical values associated with each RGB color. The corresponding site maps are then decoded by nearest-color matching in RGB space.

For an isotropic receiver, the recovered electric-field strength and received power satisfy:

$$E_{\text{dB}\mu\text{V/m}} = P_{\text{r,dBm}} + 77.2 + 20 \log_{10}(f_t/\text{MHz}), \quad (\text{S1})$$

where f_t is the carrier frequency of the corresponding configuration. Across 72 sampled configurations, the recovered values follow this relation to within 0.02 dB with unit slope, indicating that one recovered value unit corresponds to approximately 1 dB. The recovered targets are divided by 255 for training and evaluation. As an additional consistency check, the pixelwise maximum of the decoded site maps differs from the independently decoded aggregate best-server coverage map by only 0.64 value units on average.

EMF exposure example. DiReM additionally provides co-registered electric-field-strength maps. Fig. S-1 shows a representative downtown Toronto tile under the 10 dBm transmit-power setting. The peak value of 0.14 V/m corresponds to approximately 0.23% of the 61 V/m ICNIRP 2020 reference field-strength level applicable over this frequency range [1].

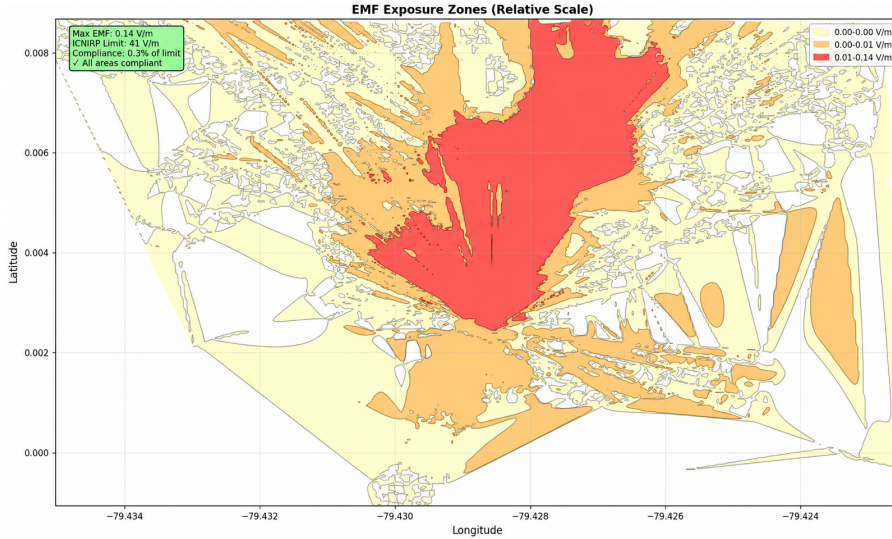


Fig. S-1. Representative DiReM electric-field-strength map for a downtown Toronto tile under the 10 dBm transmit-power setting. The map is co-registered with the corresponding radio-coverage products and assumes an isotropic receiving antenna.

Registration. Pixel coordinates follow from the geographic bounds of the CSV:

$$x_t = \frac{\text{lon}_t - \text{lon}_0}{\text{lon}_1 - \text{lon}_0} (W - 1), \quad y_t = \frac{\text{lat}_1 - \text{lat}_t}{\text{lat}_1 - \text{lat}_0} (H - 1), \quad (\text{S2})$$

where $(\text{lon}_t, \text{lat}_t)$ denotes the transmitter location, $[\text{lon}_0, \text{lon}_1] \times [\text{lat}_0, \text{lat}_1]$ defines the tile extent, and $W \times H$ is the native raster size. The coordinates are rescaled together with the maps to 256×256 . The resulting transmitter locations agree with

the centroids of independently rendered base-station markers to within approximately five native pixels (≈ 8 m), i.e., within the marker-rendering width.

Conditioning channels. The spatial conditioning tensor contains three channels. The first is the cartographic basemap $\mathbf{B}_r$, which provides the available environmental context, including roads, building outlines, land-use shading, railways, and map labels. The second encodes transmitter position as:

$$C_t^{(1)}(x, y) = \exp\left(-\frac{(x - x_t)^2 + (y - y_t)^2}{2\sigma^2}\right), \quad (\text{S3})$$

with $\sigma = 6$ native pixels, corresponding to approximately two pixels after resizing to 256×256 .

The third channel encodes antenna directivity. From the attenuation pattern $A_{a_t}(\theta)$ of antenna type a_t , we define:

$$G_{a_t}(\theta) = 10^{-A_{a_t}(\theta)/10}. \quad (\text{S4})$$

For each pixel, the angle relative to the transmitter boresight is:

$$\theta_t(x, y) = [\text{atan2}(x - x_t, y_t - y) - \phi_t] \bmod 360^\circ, \quad (\text{S5})$$

where ϕ_t is the recorded antenna azimuth. The directional channel is then:

$$C_t^{(2)}(x, y) = \alpha(P_t) G_{a_t}(\theta_t(x, y)) \rho(d_t(x, y)), \quad (\text{S6})$$

where $d_t(x, y)$ is the transmitter-to-pixel distance and

$$\alpha(P) = \min(\max(P/100, 0), 1), \quad \rho(d) = \exp\left[-\frac{d}{0.35 \max(W, H)}\right]. \quad (\text{S7})$$

Here, P is expressed in dBm. All spatial channels are generated at the native resolution and resized to 256×256 .

Transmit power and carrier frequency are additionally supplied as global scalar conditions,

$$\mathbf{v}_t = \left[\frac{P_t - 50}{50}, \frac{f_t - 2000}{500} \right]. \quad (\text{S8})$$

This scalar pathway preserves power and frequency information independently of the spatial encoding, particularly because $\alpha(P)$ saturates for $P \geq 100$ dBm.

S-II. METHODS AND PROTOCOL

We benchmark four representative learning-based methods spanning convolutional regression, state-space modeling, adversarial learning, and latent diffusion. The original architectures are retained as closely as possible and modified only where required to support the DiReM conditioning interface and target representation. All models use the three-channel spatial condition described in Sec. S-I. RadioDiff and the FiLM variant of RME-GAN additionally receive the normalized transmit-power and carrier-frequency scalars.

RadioUNet. We evaluate both the one-stage and two-stage variants of RadioUNet [2]. For DiReM, the input stem is expanded to three channels to accommodate the environmental, transmitter-location, and antenna-directivity inputs; the remaining architecture is unchanged. Each stage is trained with pixelwise ℓ_2 loss using Adam with an initial learning rate of 10^{-4} , reduced by a factor of 10 every 30 epochs. Training uses batch size 15 for 50 epochs per stage with targets normalized to $[0, 1]$. For the two-stage model, the second U-Net is initialized from the final weights of the first stage and learns to refine its prediction. Best-validation and final-epoch checkpoints are retained so that checkpoint selection can be verified consistently across experiments.

UVM-Net. UVM-Net [3] is adapted to accept the same three-channel spatial condition and to produce a single-channel radio map. Because the original image-restoration formulation predicts a residual with respect to its input, the global input-output residual connection is removed. The model is trained with an ℓ_1 objective using AdamW, batch size 8, a five-epoch learning-rate warm-up followed by cosine decay, bf16 automatic mixed precision, and gradient clipping at 1.0. We use an initial learning rate of 5×10^{-5} for 50 epochs. The larger 2×10^{-4} learning rate used in the original batch-one restoration setting was unstable under the DiReM training configuration.

RME-GAN. RME-GAN [4] uses a RadioUNet-style generator together with a PatchGAN discriminator. Its original formulation additionally relies on sparse received-signal measurements. Because DiReM is evaluated in a measurement-free setting, this input pathway and the associated gradient-similarity term are removed, while the adversarial, reconstruction, and total-variation losses retain their original weights. The resulting generator has 13,275,315 parameters and uses the same three-channel spatial condition as RadioUNet. Training uses Adam with an initial learning rate of 10^{-3} , reduced by a factor of 10 every 25 epochs, batch size 12, and 50 epochs. Targets are scaled by 256 during training, following the original implementation, and restored to the common $[0, 1]$ range before evaluation.

We additionally evaluate RME-GAN+FiLM, in which transmit power and carrier frequency are embedded by a shared two-layer MLP. Zero-initialized channel-wise modulation heads are added to the generator blocks, introducing 565,404 additional

parameters (4.3% of the generator size). Their zero initialization ensures that scalar conditioning is introduced without changing the initial behavior of the underlying generator.

RadioDiff. RadioDiff [5] performs conditional generation in a learned latent space. Its first-stage autoencoder is retrained on the decoded DiReM targets using three latent channels at 64×64 resolution for 150k optimization steps. The resulting reconstruction MSE is 2.0×10^{-4} , and the latent scaling factor is recomputed from the training-set latent standard deviation (≈ 0.56).

The latent diffusion model is then trained for 50k optimization steps using batch size 8 with four-step gradient accumulation, learning rate 5×10^{-5} , and exponential moving averaging. The three-channel spatial condition is processed by RadioDiff’s conditioning encoder, while transmit power and carrier frequency are embedded by a two-layer MLP and added to the timestep embedding. For the DiReM benchmark, inference uses the single-step decoupled sampler adopted in the main paper, after which the generated latent is decoded to a 256×256 radio map.

Evaluation controls. All per-image metrics are computed using a common evaluation implementation with a fixed dynamic range of 1.0. The metric-library version (torchmetrics 0.11.4) is recorded with the evaluation outputs to ensure consistency across software environments. Inference latency is measured at batch size one using CUDA events around the model forward pass. Three warm-up passes are performed before timing; the random state is restored afterward so that warm-up does not affect stochastic predictions. Peak GPU memory is reported as the maximum allocated memory during inference. Parameter counts include the generator for the convolutional and adversarial models and the complete inference pipeline for RadioDiff.

Verification of saved predictions. Metrics requiring model outputs are recomputed from the saved predictions as an additional consistency check. For each evaluated image, the RMSE obtained from the stored prediction is compared with the RMSE recorded during the original evaluation. Because predictions are stored with 8-bit precision, quantization can change an individual pixel by at most half a value unit. RadioDiff predictions are compared with the corresponding stochastic run that generated the saved outputs.

S-III. METRIC DEFINITIONS

Reconstruction metrics. All reconstruction metrics are computed on targets and predictions scaled to $[0, 1]$. For a target map $\mathbf{Y}$ and prediction $\hat{\mathbf{Y}}$, we report normalized mean-squared error (NMSE), root-mean-squared error (RMSE), peak signal-to-noise ratio (PSNR), and structural similarity index (SSIM). Because one recovered target value corresponds approximately to 1 dB, the RMSE in physical value units is

$$\text{RMSE}_{\text{dB}} = 255 \text{RMSE}_{[0,1]}, \quad (\text{S9})$$

and the corresponding PSNR can be written as

$$\text{PSNR} = 20 \log_{10} \left(\frac{255}{\text{RMSE}_{\text{dB}}} \right). \quad (\text{S10})$$

Variance-normalized error. Because the spatial dynamic range varies substantially among DiReM maps, we additionally report variance-normalized mean-squared error,

$$\text{vNMSE} = \frac{\text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y})}{\text{var}(\mathbf{Y})}. \quad (\text{S11})$$

Equivalently,

$$\text{vNMSE} = \text{NMSE} \frac{\overline{Y^2}}{\text{var}(\mathbf{Y})}. \quad (\text{S12})$$

The 85 test maps with target standard deviation below 0.005 are excluded from vNMSE because their near-zero spatial variance makes this normalization unstable, leaving 15,026 maps for vNMSE evaluation.

Error decomposition. For a prediction $\hat{\mathbf{Y}}$ and target $\mathbf{Y}$, the MSE admits the exact decomposition

$$\text{MSE} = (\mu_{\hat{\mathbf{Y}}} - \mu_{\mathbf{Y}})^2 + (\sigma_{\hat{\mathbf{Y}}} - \sigma_{\mathbf{Y}})^2 + 2\sigma_{\hat{\mathbf{Y}}}\sigma_{\mathbf{Y}}(1 - r), \quad (\text{S13})$$

where μ and σ denote the spatial mean and standard deviation of a map, and r is the Pearson correlation coefficient between prediction and target. The three terms respectively represent level bias, dispersion mismatch, and spatial decorrelation. In the analysis below, each term is reported as a fraction of the per-map MSE and then summarized by its median over the test set. Consequently, the three reported median shares need not sum to 100%.

Label-noise floor. To estimate the error introduced by colormap inversion, we evaluate a stratified sample of 140 configuration folders from 12 geographic areas, with four configurations sampled per area and transmit-power class. For each folder, the decoded aggregate coverage map is compared with its numerical CSV export at 256×256 resolution, and the resulting MSE is normalized by the variance of the corresponding site-map targets. Of the sampled folders, 126 from 11 areas contain site maps represented in the benchmark.

The resulting median proxy vNMSE floors are 0.666 at 10 dBm, 0.516 at 50 dBm, and 0.435 for the deployment-derived licensed-power setting, with interquartile ranges of 0.48–1.46, 0.17–0.71, and 0.22–0.66, respectively. Relative to the best

TABLE S-I

PER-CLASS COMPARISON OF THE TWO TARGET REPRESENTATIONS. ENTRIES REPORT MEDIAN vNMSE ON THE TEST SPLIT AS GRAYSCALE / RGB, WITH THE PERCENTAGE OF PAIRED TEST MAPS ON WHICH THE RGB MODEL OUTPERFORMS THE GRAYSCALE MODEL OF THE SAME ARCHITECTURE SHOWN IN BRACKETS. RADIOUNET DENOTES THE ONE-STAGE MODEL. THE 85 TEST MAPS WITH TARGET STANDARD DEVIATION BELOW 0.005 ARE EXCLUDED, LEAVING 15,026 MAPS FOR vNMSE EVALUATION.

Model	10 dBm	50 dBm	Licensed
RadioDiff	0.893 / 1.262 [20%]	1.021 / 0.764 [66%]	1.219 / 0.867 [64%]
UVM-Net	1.135 / 1.639 [21%]	0.900 / 1.019 [36%]	1.327 / 1.215 [54%]
RadioUNet	1.265 / 1.272 [52%]	0.991 / 0.911 [60%]	1.450 / 0.975 [74%]
RME-GAN	1.319 / 1.594 [38%]	1.114 / 1.202 [45%]	1.633 / 1.259 [60%]
Label-noise floor	0.666	0.516	0.435

TABLE S-II

ERROR DECOMPOSITION ON THE FULL TEST SPLIT USING SAVED PREDICTIONS. ENTRIES REPORT THE MEDIAN PER-MAP SHARE OF MSE DUE TO LEVEL BIAS, DISPERSION MISMATCH, AND SPATIAL DECORRELATION; MEDIAN SPATIAL r^2 ; THE MEDIAN RATIO OF PREDICTED TO TARGET SPATIAL STANDARD DEVIATION; AND THE CORRELATION BETWEEN PREDICTED AND TARGET MAP MEANS. THE THREE MSE-COMPONENT SHARES ARE MEDIANS OF PER-MAP QUANTITIES AND THEREFORE NEED NOT SUM TO 100%. RADIOUNET DENOTES THE ONE-STAGE MODEL.

Model	Target	Bias	Disp.	Decorr.	r^2	$\sigma_{\tilde{Y}}/\sigma_Y$	$\rho(\tilde{Y}, \bar{Y})$
RadioDiff	Gray	34.8%	16.1%	39.9%	0.455	0.66	0.901
UVM-Net	Gray	17.8%	8.3%	62.7%	0.238	0.66	0.898
RadioUNet (1-st.)	Gray	17.7%	8.0%	64.6%	0.218	0.78	0.937
RME-GAN	Gray	22.7%	8.1%	60.2%	0.189	0.74	0.939
RME-GAN+FiLM	Gray	34.7%	23.3%	24.6%	0.131	0.51	0.934
RadioDiff	RGB	4.2%	1.4%	92.6%	0.351	1.05	0.997
UVM-Net	RGB	6.5%	2.0%	90.1%	0.187	1.01	0.995
RadioUNet (1-st.)	RGB	4.4%	1.2%	93.1%	0.287	1.06	0.997
RME-GAN	RGB	7.4%	1.7%	88.7%	0.177	1.08	0.995

observed median vNMSE in each power class, these values correspond to approximately 75%, 68%, and 50%, respectively, indicating that target-recovery error can constitute a substantial fraction of the observed reconstruction error. The estimated floor also varies across geographic areas, with area-level medians ranging from 0.31 to 1.71. This analysis should be interpreted as a proxy for supervision noise rather than a direct error decomposition: independent numerical per-site ground truth is unavailable, and the estimate assumes that site maps inherit the decoding error of the aggregate coverage map because they share the same configuration-specific palette and cartographic basemap.

S-IV. OUTPUT REPRESENTATION

Setup. We evaluate both scalar grayscale targets and the original colormapped RGB representation. For the RGB track, the prediction head is changed from one to three output channels while the remaining model configuration is retained. RadioDiff’s first-stage autoencoder is likewise retrained for three-channel targets using the otherwise identical training configuration. RGB predictions are converted back to scalar values using nearest-color decoding with the configuration-specific lookup table described in Sec. S-I and are then scored against the corresponding grayscale target. Consequently, the comparison below measures reconstruction accuracy in the common scalar target space rather than error directly in RGB space.

The effect of the output representation depends on both transmit-power setting and architecture. At 10 dBm, the RGB representation increases the median vNMSE for all four models. At 50 dBm, it improves RadioDiff and RadioUNet but not UVM-Net or RME-GAN. Under the licensed-power setting, RGB improves the median vNMSE for all four architectures, with paired-map win rates ranging from 54% to 74%. Thus, the representation change does not provide a uniform gain across operating conditions; its strongest and most consistent benefit is observed under the heterogeneous licensed-power setting.

For RadioDiff, the paired grayscale/RGB comparison uses the batch-8 run for which per-image predictions were stored, ensuring that the representation comparison and paired win rates are computed from matched saved outputs. RadioDiff is stochastic, and the corresponding batch-one grayscale vNMSE values differ from these values by at most 0.01.

Table S-II shows that the two output representations produce markedly different error structures. For the grayscale models, $\sigma_{\tilde{Y}}/\sigma_Y$ ranges from 0.51 to 0.78, indicating systematically reduced spatial variation relative to the targets. Relative to the RGB rows, the grayscale models also exhibit substantially larger level-bias and dispersion-mismatch contributions.

For the RGB models, spatial decorrelation accounts for 88.7–93.1% of the median per-map error decomposition, while $\sigma_{\tilde{Y}}/\sigma_Y$ remains close to unity (1.01–1.08) and the correlation between predicted and target map means is 0.995–0.997. Thus,

the RGB results are characterized by well-matched map-level mean and dispersion, while the remaining error is dominated by spatial mismatch. In contrast, the grayscale models exhibit a stronger mixture of level, dispersion, and spatial errors.

S-V. RADIOBRIDGE IMPLEMENTATION DETAILS

Datasets and protocol. RadioBridge is evaluated on RadioMapSeer and DiReM. For RadioMapSeer, we use the dominant-path-model (DPM) simulations and the reference RadioUNet partition [2], [6]. The 700 city maps are shuffled with seed 42 into 501 training, 100 validation, and 99 test maps. With 80 transmitter locations per map, the test partition contains 7,920 transmitter–map instances. Following the reference preprocessing, normalized gains below 0.2 are clipped to 0.2 and the resulting targets are rescaled to $[0, 1]$. The static radio-map (SRM) setting uses building and transmitter-location masks, while the dynamic radio-map (DRM) setting additionally includes the vehicle layer and the corresponding vehicle-aware propagation targets.

For DiReM, we use the patch-cluster-disjoint benchmark partition described in the main paper, containing 15,111 test instances. The split is disjoint at the patch-cluster level but does not enforce transmitter-site disjointness across training, validation, and test sets. Both grayscale and RGB target representations are evaluated. The deterministic prior and the refiner receive the same three-channel spatial condition and normalized power–frequency scalars described in Sec. S-I.

Deterministic prior. The deterministic prior g_ψ is implemented as a ConvNeXt-style U-Net. For RadioMapSeer, its encoder widths are (64, 128, 256, 512) with depths (2, 2, 6, 2), giving approximately 10.9M parameters. The prior is trained for 60 epochs using AdamW with an initial learning rate of 10^{-3} , weight decay 10^{-4} , and batch size 64. The learning rate follows cosine decay with a 10% warm-up period. Training uses mixed precision and an exponential moving average (EMA) with decay 0.999. The EMA checkpoint with the lowest validation loss is retained.

The same prior architecture and optimization configuration are used for DiReM, with the input interface extended to the spatial and scalar transmitter conditions defined in Sec. S-I.

Refiner architecture and training. The refinement network is based on the DiffUNet architecture used by RadioFlow [7]. We evaluate two capacities. The *lite* model uses widths (32, 32, 64, 128, 256, 32) and contains approximately 4.0M parameters, whereas the *large* model uses widths (128, 128, 256, 512, 1024, 128) and contains approximately 54.1M parameters.

The refiner is trained for 120 epochs using AdamW with learning rate 2×10^{-4} , weight decay 10^{-5} , batch size 64, cosine learning-rate decay with a 5% warm-up period, mixed precision, and EMA decay 0.999. The pretrained prior remains frozen throughout refiner training so that the anchor distribution does not change while the refinement model is being optimized.

Refiner checkpoints are selected according to the one-step validation NMSE of the complete prior–refiner pipeline, evaluated on a fixed set of 512 validation instances. Unless stated otherwise in the main paper, the bridge interpolation uses

$$\lambda = 0.5, \quad \sigma_b = 0.05. \quad (\text{S14})$$

The large refiner is used for the primary RadioBridge results, while the lite configuration is retained for the capacity ablation.

Noise-anchored controls. Noise-anchored comparison models use the large DiffUNet backbone. Conditional flow matching (CFM) follows the public RadioFlow implementation [7] and is trained for 800 epochs. Increasing the training duration from 500 to 800 epochs changes the one-step NMSE from 0.0116 to 0.0118, indicating that additional training does not materially improve the one-step result. Mean flow [8] is evaluated both from scratch and by fine-tuning the converged CFM checkpoint. For the CFM-initialized variant, we use a learning rate of 2×10^{-5} . The CFM-initialized model is the variant reported in the main comparison. These models are used as controlled noise-anchored comparisons under the same evaluation pipeline rather than as claims of reproducing the published optimum.

Computational cost and NFE. Network function evaluations (NFE) count evaluations of the deterministic prior and the denoising or refinement network. A RadioBridge prediction with K refinement steps therefore uses

$$\text{NFE} = 1 + K, \quad (\text{S15})$$

where the additional evaluation corresponds to the prior. Noise-anchored models use K predictor evaluations. Condition encoders and, where applicable, autoencoder decoders are executed once per map and are included in the measured wall-clock latency but not in the NFE count. Counting the prior as one network evaluation therefore gives a conservative accounting of RadioBridge sampling cost.

RadioMapSeer latency is measured at batch size one on an NVIDIA H100 using CUDA synchronization. RadioBridge and the noise-anchored controls use 10 warm-up runs followed by 50 timed runs. For DiReM, batch-one latency is measured on an NVIDIA A100 using the benchmark evaluation pipeline. Reported RadioBridge parameter counts and latency include the deterministic prior.

RadioMapSeer evaluation. For RadioMapSeer, all reconstruction metrics are computed independently for each test map and then averaged over the 7,920 test instances. NMSE normalizes each map’s MSE by the mean squared thresholded target, PSNR uses a dynamic range of $L = 1$, and SSIM uses an 11×11 Gaussian window with standard deviation 1.5; RMSE is also computed per map. This aggregation is independent of the inference batch size and is applied consistently to the primary comparison, ablations, and prior-perturbation experiments. Latency is measured separately at batch size one.

TABLE S-III

SENSITIVITY OF LICENSED-POWER RESULTS TO REGISTRY VALUES ABOVE 60 dBm. MEDIAN NMSE ($\times 10^{-3}$) AND, IN BRACKETS, MEDIAN vNMSE PER MAP ARE REPORTED FOR THE 5,052 LICENSED-POWER TEST MAPS. OF THESE, 3,456 (68%) EXCEED 60 dBm, INCLUDING 3,120 AT THE SINGLE REGISTRY VALUE 159.2 dBm. THE LAST TWO COLUMNS REPORT MEDIAN NMSE OVER THE FULL TEST SPLIT WITH AND WITHOUT MAPS ABOVE 60 dBm.

Model	Licensed-power maps			Whole test split	
	All	≤ 60 dBm	> 60 dBm	All	Excl. > 60 dBm
RadioDiff	16.34 [1.209]	49.60 [1.304]	6.72 [1.165]	19.59	22.02
UVM-Net	21.04 [1.327]	55.76 [1.490]	7.44 [1.260]	21.42	24.79
RadioUNet, two-stage	25.65 [1.583]	70.37 [1.674]	7.95 [1.532]	24.84	27.24
RadioUNet, one-stage	26.91 [1.450]	60.55 [1.654]	8.67 [1.410]	25.38	28.06
RME-GAN	28.36 [1.633]	50.48 [1.585]	11.70 [1.664]	27.67	30.63
RME-GAN + FiLM	26.98 [3.123]	116.50 [3.480]	15.24 [3.006]	27.89	31.03
RadioBridge prior	19.63 [1.551]	78.08 [1.749]	4.67 [1.475]	24.73	28.83
RadioBridge, $K = 1$	21.73 [1.604]	104.27 [2.247]	4.95 [1.422]	24.07	28.72

Reproduction scope. The RadioDiff and CFM implementations used as controls should be interpreted as in-protocol reproductions rather than reproductions of the published optima. Our RadioDiff implementation is trained for 50k optimization steps and sampled with 50 reverse steps on RadioMapSeer, substantially below the training and sampling budget reported in the original work [5]. The CFM baseline follows the public RadioFlow architecture and training recipe [7], but our reproduction does not recover the published RadioFlow accuracy. We therefore use these models only for controlled comparisons under the common preprocessing, evaluation, and timing protocol adopted in this work.

Published RadioMapSeer numbers are not directly combined with our measured results because target preprocessing, metric aggregation, and latency measurement procedures differ across studies. In particular, our RadioMapSeer evaluation follows the thresholded-target loader described above, whereas several published results use unthresholded targets. All quantitative comparisons reported for RadioBridge in the main paper are therefore based on models evaluated under the same local protocol.

Sensitivity to high licensed registry values. The deployment-derived licensed-power setting contains a substantial number of registry values above 60 dBm. Of the 5,052 licensed-power test maps, 3,456 (68%) have values above 60 dBm, and 3,120 of these use the single registry value 159.2 dBm. Because these values are taken directly from the registry and are not interpreted as calibrated transmitter powers or EIRP, we evaluate whether they materially affect the reported conclusions by repeating the analysis after excluding all maps above 60 dBm.

Excluding maps above 60 dBm preserves RadioDiff as the best-performing model on the licensed-power subset in both NMSE and vNMSE. Absolute NMSE, however, increases by approximately $1.8\text{--}4.8\times$ across models when these maps are removed. The excluded maps have high field levels, which increase the mean-squared target term in the NMSE denominator and make their normalized errors comparatively small; vNMSE is substantially less sensitive for most models.

The effect is particularly pronounced for models that explicitly receive transmit power as a scalar condition. For example, the RadioBridge prior achieves 4.67×10^{-3} NMSE on the > 60 dBm subset but 78.08×10^{-3} after restricting the licensed-power maps to ≤ 60 dBm. On the full test split, the prior is comparable to the two-stage RadioUNet (24.73 versus 24.84×10^{-3} NMSE), but after excluding the > 60 dBm maps it trails RadioUNet (28.83 versus 27.24×10^{-3}). We therefore treat the licensed-power configuration as a registry-derived stress test rather than a physically calibrated power benchmark, while the fixed 10 and 50 dBm settings provide the controlled power comparisons.

S-VI. ROBUSTNESS TO PRIOR PERTURBATIONS

Protocol. To test whether bridge-state training improves robustness when the anchor is imperfect, we perturb the deterministic prior at test time on RadioMapSeer SRM without retraining either refiner. We consider Gaussian blur with $\sigma \in \{2, 4\}$ pixels, additive Gaussian noise with standard deviation $\sigma_n \in \{0.02, 0.05\}$ in the normalized $[0, 1]$ target space, and a multiplicative gain error $\tilde{\mathbf{p}} = 0.9\mathbf{p}$. The matched cascade and RadioBridge use the same large refiner backbone and are evaluated at $K \in \{1, 4\}$; the $K = 4$ results use deterministic re-anchoring. NMSE is computed independently for each map and averaged over all 7,920 test instances. All other evaluation settings follow Sec. S-V.

TABLE S-IV

ROBUSTNESS TO A DEGRADED PRIOR ON RADIOMAPSEER SRM. THE PRIOR OUTPUT IS DEGRADED AT TEST TIME AND BOTH REFINERS ARE APPLIED UNCHANGED. NMSE ($\times 10^{-3}$) IS COMPUTED PER MAP AND AVERAGED OVER THE 7,920 TEST INSTANCES. THE $K = 4$ RESULTS USE DETERMINISTIC RE-ANCHORING.

Prior degradation	Prior	Cascade		RadioBridge	
		$K = 1$	$K = 4$	$K = 1$	$K = 4$
None	6.85	5.64	5.96	5.66	5.77
Blur, $\sigma = 2$ px	31.15	24.06	21.47	20.70	20.37
Blur, $\sigma = 4$ px	60.82	47.16	41.04	41.84	36.53
Noise, $\sigma_n = 0.02$	14.25	9.93	9.10	6.10	6.20
Noise, $\sigma_n = 0.05$	52.25	27.93	22.58	9.09	8.65
Gain, $\times 0.9$	16.29	6.50	6.14	6.24	6.07

Results. With the clean prior, the matched cascade and RadioBridge achieve essentially the same one-step accuracy (5.64 versus 5.66×10^{-3} NMSE). Under every prior degradation, however, RadioBridge is more accurate than the cascade at the same refinement depth. The largest difference occurs under additive noise: for $\sigma_n = 0.05$, one RadioBridge step reaches 9.09×10^{-3} NMSE, compared with 27.93×10^{-3} for the cascade. This behavior is consistent with the cascade being trained only on the exact prior output, whereas bridge-state training exposes the refiner to intermediate states between the prior and target. Additional refinement can also become useful when the anchor is degraded; for example, under $\sigma = 4$ px blur, deterministic $K = 4$ refinement reduces RadioBridge NMSE from 41.84 to 36.53×10^{-3} , while additional steps do not improve the clean-prior result. These synthetic perturbations probe sensitivity to errors in the anchor rather than performance with a naturally weaker prior.

S-VII. ADDITIONAL RADIOMAPSEER BASELINE CHECK

As an additional cross-dataset check, we evaluate the three non-diffusion benchmark architectures on RadioMapSeer using the same local evaluation harness employed for the RadioBridge experiments. This experiment is not intended to reproduce the published absolute numbers of the original methods; rather, it verifies their relative behavior under a common preprocessing and evaluation protocol.

TABLE S-V

RADIOMAPSEER SRM RESULTS FOR THE THREE NON-DIFFUSION BASELINE ARCHITECTURES UNDER THE LOCAL EVALUATION PROTOCOL. THE EXPERIMENT USES THRESHOLDED DPM TARGETS, 7,920 TRANSMITTER-MAP TEST INSTANCES, BATCH-ONE EVALUATION, AND AN NVIDIA H100. GPU DENOTES PEAK ALLOCATED INFERENCE MEMORY.

Method	NMSE	RMSE	SSIM	PSNR (dB)	ms/map	GPU (MB)
RadioUNet, two-stage	0.01068	0.01961	0.9240	34.45	2.56	135.5
RME-GAN	0.01373	0.02228	0.9123	33.29	2.51	136.0
UVM-Net	0.01492	0.02381	0.8995	32.62	26.41	4165.0

Under this common protocol, RadioUNet, RME-GAN, and UVM-Net retain the same relative ordering in PSNR as in the corresponding RadioMapSeer comparison reported in the main paper. The two-stage RadioUNet reaches 34.45 dB, RME-GAN 33.29 dB, and UVM-Net 32.62 dB. RadioUNet and RME-GAN require approximately 136 MB of peak allocated inference memory, whereas UVM-Net requires approximately 4.17 GB under the same batch-one evaluation.

The one-stage RadioUNet reaches 33.87 dB PSNR under the same protocol, corresponding to a 0.58 dB gain from the second stage. These results are included only as a cross-dataset implementation check. Differences between RadioMapSeer and DiReM should not be attributed to a single factor because the datasets differ in geographic partitioning, transmitter configurations, conditioning information, and target construction.

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